KENDRIYA VIDYALAYA SANGATHAN
BHUBANESWAR REGION
1st PRE BOARD EXAMINATION (2024-25)
CLASS X
Science (086)

Max. Marks: 80 **Time Allowed: 3 hours**

General Instructions:

1. All *questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.*

2. Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.

3. Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.

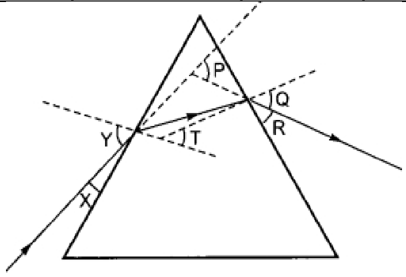
4. Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.

5. Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.

6. Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.

Section-A

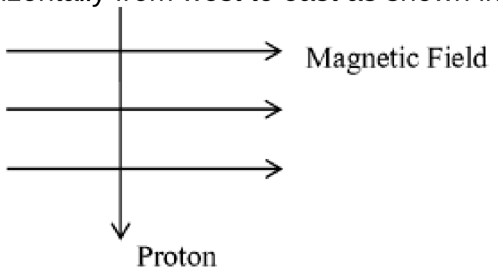
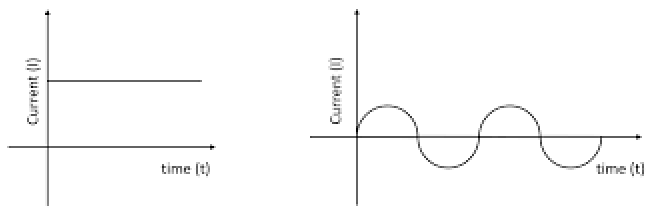
Question 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions.

1.	Gustatory receptors are able to detect:- a) Touch b) Smell c) light d) Taste	**1**
2.	Solid calcium oxide reacts vigorously with water to form calcium hydroxide accompanied by the liberation of heat. From the information given above it may be concluded that this reaction is:- a) Endothermic and pH of the solution is more than 7. b) Exothermic and pH of the solution is 7. c) Endothermic and pH of the solution is 7. d) Exothermic and pH of the solution is more than 7.	**1**
3.	The strength of magnetic field inside a long current carrying straight solenoid is a) More at the ends than at the centre b) Minimum at the middle c) Same at all points d) Found to increase from one end to the other	**1**
4.	Amongst the following group of materials the group which contains only biodegradable materials is:- a) Polythene, Soap, Wool c) Plastic, Leather, Grass b) Glass, Aluminium, Neem Leaves d) Wood, Leather , Paper	**1**
5.	The image formed by a concave mirror of focal length 25 cm is real and of magnification -1. In this case the distance between the objet and its own image is:- a) 50 cm b) zero c) 25 cm d) 100 cm	**1**
6.	 In the following diagram, the path of a ray of light passing through the glass prism is shown. In this figure the angle of incidence, the angle of emergence and the angle of deviation are-: a) X, R and Q b) X, R and P	**1**

	c) Y, Q and P d) Y, T and Q	
7.	A hydrocarbon which contains two C-C single bonds and one C=C double bond is: a) Ethene b) Benzene c) Butene d) Propene	**1**
8.	Decomposers are not included in the food chain. The correct reason for the same is because decomposers:- a) Act at every tropic level of the food chain. b) Do not breakdown organic compounds. c) Convert organic material to inorganic forms. d) Release enzymes outside their body to convert organic material to inorganic forms.	**1**
9.	An element X reacts with O_2 to give a compound with a high melting point. This compound is also soluble in water. The element 'X' is likely to be :- a) Iron b) Carbon c) calcium d) silicon	**1**
10.	Solder is an alloy of : a) Cu and Zn b) Pb and Sn c) Pb, Sn and Zn d) Al and Sn	**1**
11.	Which of the following statements about the given reaction are correct? $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$ i) Iron metal is getting oxidized. ii) Water is getting reduced. iii) Water is acting as a reducing agent. iv) Water is acting as an oxidising agent. a) i, ii and iii b) i , ii and iv c) iii and iv d) ii and iv	**1**
12.	The maximum resistance of a network of five identical resisters of 1/5 ohm each can be :- a) 1 ohm b) 0.5 ohm c) 0.25 ohm d) 0.1 ohm	**1**
13.	In plants, A stomata closes when: i) It needs CO2 for photosynthesis. ii) It does not need CO2 for photosynthesis. iii) Water flows into the guard cells. iv) Water flows out of the guard cells. a) Only ii b) ii and iii c) i and iv d) ii and iv	**1**
14.	Walking in a straight line and riding a bicycle are the activities which are possible due to a part of the brain. Choose the correct location and name of this part from the given table:- <table><tr><td></td><td>Part of brain</td><td>Name</td></tr><tr><td>a</td><td>Fore brain</td><td>Cerebrum</td></tr><tr><td>b</td><td>Mid brain</td><td>Hypothalamus</td></tr><tr><td>c</td><td>Hind brain</td><td>Cerebellum</td></tr><tr><td>d</td><td>Hind brain</td><td>Medulla</td></tr></table>	**1**
15.	Select from the following the conditions responsible for the rapid spread of bread mould on a slice of bread:- i) Formation of large number of spores. ii) Presence of moisture and nutrients in bread. iii) Low Temperature iv) Presence of hyphae a) i and ii b) ii and iv c) Only ii d) ii and iii	**1**
16.	If homozygous tall plant cross with heterozygous tall plant, What will be the percentage of tall and dwarf in their offspring? a) 75 tall and 25 dwarf c) 100 tall and no dwarf b) 25 tall and 75 dwarf d) 50 tall and 50 dwarf	**1**

Question No. 17 to 20 consist of two statements – **Assertion** (A) and **Reason** (R). Answer these questions by selecting the appropriate option given below:
A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.

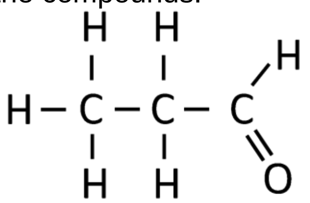
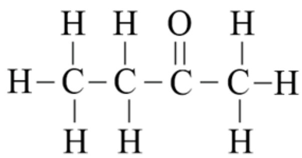
D. A is false but R is true		
17.	Assertion (A) : Magnetic field lines never intersect each other. Reason (R) : The magnetic lines arise from north pole to south pole of the magnet..	**1**
18.	Assertion (A) : Ozone (O3) is a molecule formed by the three atoms of oxygen. Reason (R) : Ultraviolet sunlight breaks apart an oxygen molecule to form two separate oxygen atoms. Then each oxygen atom combines with oxygen molecule to form ozone.	**1**
19.	Assertion (A) : In human, there is a complex respiratory system. Reason (R) : Human skin is permeable to gases.	**1**
20.	Assertion (A) :After white washing the walls, a shiny white finish on walls is obtained after two to three days. Reason (R) : Calcium oxide reacts with carbon dioxide to form calcium hydrogen carbonate which gives shiny white finish.	**1**
Section-B **Question No. 21 to 26 are very short answer questions**		
21.	*Attempt either option A or B.* A. List the role of each of the following in our digestive system:- a) Mucus b) Trypsin **OR** B. Name the following : a) The three carbon molecule that is formed due to break down of glucose during respiration. b) The nitrogenous waste that is removed from the blood in our kidneys.	**2**
22.	Translate the following statements into chemical equations and balance them. a) Hydrogen sulphide gas burns in air to give water and sulphur dioxide. b) Silver bromide on exposure to sunlight decomposes into Silver and Bromine.	**2**
23.	What is Placenta? Explain its function in humans.	**2**
24.	*Attempt either option A or B.* A. Draw a labelled ray diagram to show the path of a ray of light incident obliquely on one face of a glass slab. Show the angle of incidence, the angle of emergence and the lateral displacement. **OR** B. If the image formed by mirror for all positions of the object placed in front of it always virtual and diminished. State the type of mirror. Draw a ray diagram in support of your answer.	**2**
25.	Draw the structure of the following compounds indicating the functional group present in each a) Propanol b) Butanoic acid	**2**
26.	Why do we need to clean aquarium but not a pond? Explain.	**2**
SECTION C **Question No. 27 to 33 are short answer questions**		
27.	(a) A non metal X exists in two different forms Y and Z. Y is the hardest natural substance whereas Z is a good conductor of electricity. Identify Y and Z. (b) Show the electron dot structures of the formation of CaO and $MgCl_2$.	**3** **(1+2)**
28.	*Attempt either option A or B.* A. List the important products of the chlor-alkali process. Write one important use of each product. **OR**	**3**

	B. In one of the industrial processes used for manufacture of sodium hydroxide, a gas X is formed as by product. The gas X reacts with lime water to give a compound Y which is used as a bleaching agent in chemical industry. Identify X and Y and write the chemical reactions involved.	
29.	*<u>Attempt either option A or B.</u>* A. (a) State (i) right-hand thumb rule, and (ii) Fleming's left-hand rule. (b) Using Fleming's left-hand rule determine the direction of force experienced by a proton, which enters vertically downwards in a uniform magnetic field acting horizontally from west to east as shown in the diagram.  **OR** **B.**  (a) (b) i) Name the type of current in two cases with one difference between them. ii) What is the frequency of current in case (b) in India.	**3** **(2+1)**
30.	Write the difference between two kinds of blood vessels present in Human beings. Write minimum three points of each type	**3**
31.	State the cause of dispersion of white light by a glass prism. How did Newton using two identical glass prisms, show that white light is made of seven colours? Draw the ray diagram in support of above activity.	**3**
32.	(a) Why did Mendel choose garden pea plant? Write two reasons. (b) How do Mendel's experiment show that traits are inherited independently? Explain it.	**3** **(1+2)**
33.	(a) Draw a schematic diagram of a circuit consisting of a battery of four 1.5V cells, a 5 ohm register, 10 ohm and 15 ohm resistors in a series connection with ammeter and volt meter across 10 ohm resistors. (b) Find the electric current passing through the circuit. (c) What is the potential difference across 10 ohm resistors?	**3** **(1+1+1)**

Section-D

Question No. 34 to 36 are long answer questions.

34.	*<u>Attempt either option A or B</u>*. A. (a) Why sexual reproduction is considered advantageous over asexual reproduction? (b) What happens to different parts of the flower after fertilisation?	**5**

<table>
<tr><td></td><td>(c) Draw neat diagram of a germinating seed and label the part which:- i) Store food ii) Forms root iii) Forms shoot.
OR
B.
(a) Testes in male human reproductive part perform dual functions. Explain it.
(b) Describe the role of prostate gland and seminal vesicle in human male reproductive system.
(c) Name the surgical method in male human being to prevent the pregnancy and write its mechanism.</td><td></td></tr>
<tr><td>35.</td><td><u>Attempt either option A or B.</u>
A.
(a) Upper part of a convex lens is covered with a black paper.draw a ray diagram to show the formation of image of an object placed at 2F from such a lens. Mention the position and nature of the image formed. State the observable difference in the image obtained if the lens is uncovered. Give reason to justify your answer.
(b) An object is placed at a distane of 30 cm from the optical centre of a concave lens of focal length 15 cm. Use lens formula to determine the distance of the image from the optical centre of the lens.
OR
B.
(a) A student has focused the image of an object on a white screen of concave mirror. The length of the object is 10 cm, Focal length of the mirror is 12cm and the distance of the object from the mirror is 18 cm. Calculate the following:-
i) Distance of the image from the mirror.
ii) Length of the image.
(b) If the distance between the mirror and the object is reduced to 10 cm, then what would be observed on screen? Draw a ray diagram to justify your answer for this situation.</td><td>5</td></tr>
<tr><td>36.</td><td><u>Attempt either option A or B.</u>
A.
(a) Identify the functional groups present in the following carbon compounds and name the compounds.
i)  ii) 
(b) How can you get back alcohol from ester? Write the name of the reaction with its equation.
(c) Name the reaction which is used to convert vegetable oil to vanaspati ghee in industry. Write the equation.
OR
B.
(a) Describe the cleansing action of soap with proper diagram.
(b) Write the major three difference between soap and detergent.</td><td>5</td></tr>
<tr><td colspan="3">Section – E
Question No. 37 to 39 are case-based/data -based questions.</td></tr>
<tr><td>37.</td><td></td><td>4</td></tr>
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Compound X $\xrightarrow{+Zn}$ A + H2O (i)

Compound X $\xrightarrow{+HCl}$ B + H2O (ii)

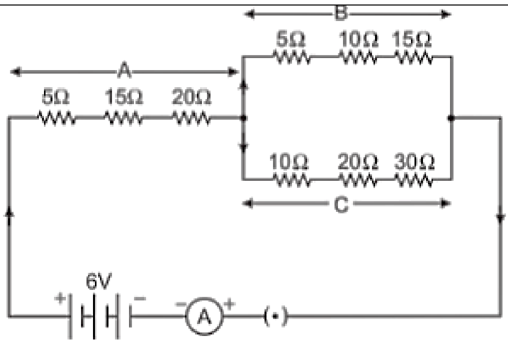
Compound X $\xrightarrow{+CH3COOH}$ C +H2O (iii)

A. Identify the compound X with its common name on the basis of the reactions given above.
B. Which of the reaction is an example of neutralization?
a) I) b) i) & ii) c) ii) & iii) d) i) & iii)

Attempt either subpart C or D.

C. Write the name and chemical formulae of A and B.
OR
D. Draw the electron dot structure of acetic acid (CH3COOH).

38. **4**



Study the above electric circuit in which the resisters are arranged in three arms A,B and C and answer the following questions.

a) Calculate the equivalent resistance of the connection of B and C arms .
b) What will be the reading show in Ammeter?

Attempt either subpart c or d.

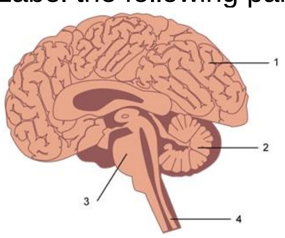
c) What would be the reading of ammeter if the arm B is removed from the circuit?
OR
d) What will be the potential difference across the A arm?

39. **4**

Spinal cord is a cylindrical structure. It begins in continuation with the medulla oblongata of brain and extends downwards upto early part of lumbar region. Internally the spinal cord possesses a narrow fluid filled cavity called central canal. Spinal cord is enclosed in vertebral column or backbone which protects it.

Attempt either subpart a or b.

a) How is the brain protected against injuries and shocks?
OR
b) Label the following parts:- Any two with their function each of the part.



c) Write one difference between reflex action and walking.

	d) Name one important endocrine gland present inside the brain which is known as master gland. Write the function of its secreting hormone.	